

Beware: It's Easy to Launch a Wireless Deauthentication Attack!

This article shows how easily someone can pull off an attack on an 802.11 Wi-Fi network. The author sets up a mock attack using a VMware virtual machine to help readers understand the process. While these attacks are more of a nuisance value, they can pose a security threat to a wireless network in the production environment.



The era of wireless connectivity gives flexibility and mobility, but also comes with security issues.

In wired connectivity, the attacker needs physical access in order to connect and attack. In the case of wireless connectivity, an attacker just needs the availability of the signal to launch an attack.

In this article, I will discuss the very serious deauthentication attack. Even your neighbour's kid could make your life miserable by launching such an attack, in which your device can get disconnected from the access point (AP).

Before proceeding further, you will need to get familiar with some of the terminology we will use in this article.

Service Set Identification (SSID): An SSID is a 32-character (maximum) alphanumeric key identifying the name of the wireless local area network (WLAN).

AP (Access Point): This is used to connect the wireless device to the wireless network.

BSSID: MAC address of the AP.

The concept behind the attack

So, why are these attacks made? Where, and typically, who attacks whom? Well, the attack can be made by a penetration tester on a company's wireless network, if the company or organisation wishes to check the robustness of its own wireless security. The tester then sends a report of the findings to the company. But some hackers mount such attacks simply to create a nuisance for users.

The connection between the clients and APs is established by the exchange of various frames, as shown in Figure 1. The communication between the client and the AP is established after probing the available wireless APs. After that, the exchange of a series of management frames, like authentication and association request frames, takes place.

Launching the deauth attack

In order to launch the attack, I used the following:

- Kali Linux as the attacker